Advanced Mathematical Techniques Worksheet 2
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The purpose of this worksheet is to give you some practice with finding Taylor series, determining their radius of convergence, determining the value of integrals by differentiating with respect to an auxiliary parameter, and working with infinite product expansions. Please do all of these problems without a calculator and clearly explain what you are doing at each step.

1. Find the first four nonzero terms in the Maclaurin series for $\frac{x^3}{e^{2x^2}-1}$. What is the radius of convergence of this expansion?

2. Find the first three nonzero terms in the Maclaurin series for $\tan\left(2x^3\right)$. What is the radius of convergence of this expansion? Use your result to determine the value of the 9th and 12th derivative of this function at zero.

3. Determine the value of the integral $\int_{-\infty}^{\infty} \frac{x^2}{\left(x^2+9\right)^3}\,dx$.

4. Determine the value of the integral $\int_{-\infty}^{\infty} \frac{x^4}{\left(x^2+1\right)^6}\,dx$.

5. Determine the value of the integral $\int_0^1 x^7 \ln^4 x\,dx$.

6. Determine the value of the integral $\int_0^1 x\sqrt{x}\,\ln^2 x\,dx$.

7. Determine the value of the integral $\int_0^\infty x^6 e^{-3x^2}\,dx$.

8. The infinite product expansion for the sine function, $\frac{\sin x}{x} = \prod_{k=1}^{\infty}\left(1-\frac{x^2}{k^2\pi^2}\right)$, can be used to determine the value of many interesting infinite products.

   (a) Determine the value of $\frac{1\cdot 3}{2\cdot 2}\cdot\frac{3\cdot 5}{4\cdot 4}\cdot\frac{5\cdot 7}{6\cdot 6}\cdot\cdots = \prod_{k=1}^{\infty}\frac{4k^2-1}{4k^2}$.

   (b) Determine the value of $\frac{2\cdot 4}{3\cdot 3}\cdot\frac{5\cdot 7}{6\cdot 6}\cdot\frac{8\cdot 10}{9\cdot 9}\cdot\cdots = \prod_{k=1}^{\infty}\frac{9k^2-1}{9k^2}$.

   (c) Explain why the infinite product $\prod_{k=1}^{\infty}\frac{k^2-1}{k^2}$ is obviously zero. Then, use a limit process with the sine function's infinite product to determine $\prod_{k=2}^{\infty}\frac{k^2-1}{k^2}$.

   (d) Use complex numbers along with the infinite product representation of the sine function to determine the value of $\prod_{k=2}^{\infty}\frac{k^2+1}{k^2}$.

9. Use complex numbers along with the infinite product expansion for the sine function to determine the value of $\prod_{k=1}^{\infty}\left(\frac{4k^2+1}{4k^2}\right) = \frac{5}{4}\cdot\frac{17}{16}\cdot\frac{37}{36}\cdot\frac{65}{64}\cdot\cdots$.